# Wang Ma

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## Summary

Ph.D. student working on trustworthy LLMs and agents (uncertainty quantification, abstention, reliable reasoning and RAG), with first-author papers at **NeurIPS 2026** and **CVPR 2026 (Highlight)**. Industry experience improving an LLM agent at Microsoft and developing black-box uncertainty quantification for LLMs at IBM Research. Seeking Summer 2027 research and applied science internships in trustworthy AI, LLM reliability, and generative AI.

## Education

**Rensselaer Polytechnic Institute (RPI)** Aug 2024 – May 2029 (Expected)
*Ph.D. in Computer & Systems Engineering (Advisor: Prof. Qiang Ji)* *Troy, NY*
**Research Interests:** LLM Uncertainty, Abstention, Reliable Reasoning & RAG, Bayesian Deep Learning, Knowledge-augmented AI

**Southern University of Science and Technology (SUSTech)** Aug 2020 – July 2024
*B.S. in Data Science and Big Data Technology (Advisor: Prof. Chao Wang)* *Shenzhen, China*

**University of California, Irvine (UCI)** Mar 2023 – July 2023
*Exchange Student (ASAP Program)* *Irvine, CA*

## Publications & Manuscripts

1. **Black-Box Uncertainty Quantification for Large Language Models via Ensemble-of-Ensembles**
**Wang Ma**, Debarun Bhattacharjya, Junkyu Lee, Nhan H Pham, Harsha Kokel, Qiang Ji.
*NeurIPS 2026.* Preliminary version at AAAI 2026 AIR-FM Workshop.
2. **Towards Knowledge-augmented Bayesian Deep Learning For Computer Vision**
**Wang Ma**, Hanjing Wang, Yufei Zhang, Darsha Udayanga, Qiang Ji.
*CVPR 2026.* ***Highlight***.
3. **Distilling Bayesian Uncertainty into a Single Forward Pass**
**Wang Ma**, Qiang Ji.
*Under review at ICLR 2027.*
4. **Causal Saliency Map For Post-hoc Model Explanation**
Hanjing Wang, **Wang Ma**, Naiyu Yin, Qiang Ji.
*Under review.*

## Industrial Experience

**Microsoft** **Redmond, WA**
*Applied Science Intern, PPT ML Team* *May 2026 – Aug 2026*
- Worked on PowerPoint's **LLM-based slide generation agent**; diagnosed and fixed table-rendering issues in agent-generated slides to improve presentation quality.
- Analyzed agent execution traces to identify wasted iterations in the **build pass**, and implemented a **deterministic runtime tracker** to improve convergence and reduce end-to-end generation latency.

**IBM Research** **Yorktown Heights, NY**
*Research Extern (Mentor: Dr. Debarun Bhattacharjya)* *May 2025 – Aug 2025*
- Developed a **perturbation-based UQ framework** for **black-box LLMs**, decomposing uncertainty without access to model weights or intermediate activations.
- Introduced an "ensemble-of-ensembles" method to disentangle **aleatoric and epistemic uncertainty**, with AUROC comparable to or better than white-box baselines; first-authored paper accepted to **NeurIPS 2026**.

## Academic Experience

**Rensselaer Polytechnic Institute (RPI)** **Troy, NY**
*Graduate Research Assistant (Advisor: Prof. Qiang Ji)* *Sept 2024 – Present*
- **Knowledge-Augmented Bayesian Deep Learning:** Proposed a two-stage framework with a **knowledge-induced prior** and **adaptive knowledge likelihood**, improving accuracy, robustness, and **OOD detection**.
- **Single-Pass Bayesian Uncertainty:** Distilled Bayesian ensemble uncertainty into a single-forward-pass student via a two-stage knowledge distillation pipeline, enabling efficient uncertainty estimation at inference time.
- **Causal Saliency Maps:** Implemented **Causal Markov Blanket (CMB)** detection algorithms using label-wise mutual information to identify causally relevant features and filter spurious correlations.
- **Semantic Uncertainty Disentanglement:** Developed a **contrastive learning framework** to disentangle semantic and nuisance uncertainties, achieving strong benchmark performance. (Technical Report)
- **Efficient Single-Model UQ:** Used **NTK theory** and **credal interval-based methods** to study practical uncertainty quantification with a single auxiliary neural network. (Technical Report)

**SUSTech** **Shenzhen, China**
*Undergraduate Researcher (Advisor: Prof. Chao Wang)* *Mar 2024 – Aug 2024*
- **Hyperspectral Image Restoration:** Built a self-supervised method combining **Implicit Neural Representations (INR)** and **Diffusion Models**.
- **Unpaired Image Denoising:** Optimized a **VAE** from unpaired data with Mutual Information maximization.

## Technical Skills

- **Programming:** Python (PyTorch, NumPy, Pandas, Scikit-learn), Java, MATLAB, LaTeX
- **Developer Tools:** Git, Docker, Linux
- **Natural Languages:** Mandarin Chinese (Native), English (Fluent), Japanese (Conversational)

## Honors & Service

- **Golden Reviewer**, International Conference on Machine Learning (ICML) 2026
- **SUSTech Excellent Undergraduate Graduation Project** (Thesis: End-to-end Unpaired Image Denoising) June 2024
- **Seminar Organizer:** AI Optimization & Theory (SUSTech); led sessions on Bayesian Optimization & Bayesian Deep Learning 2024
- Second Prize in Guangdong Province, Mathematical Contest in Modeling 2021
- **Scholarships:** SUSTech Excellent Student Scholarship (2021, 2022)